

B.TECH
(SEM V) THEORY EXAMINATION 2022-23
AUTOMOBILE ENGINES AND COMBUSTION

Time: 3 Hours**Total Marks: 100****Note: 1.** Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.**2 x 10 = 20**

- (a) What are the properties of LPG as a SI engine fuel?
- (b) Calorific value of the fuel.
- (c) Draw P-V and T-S diagrams for dual cycle.
- (d) Why 2 strokes engines are not popular in 2 wheelers in current scenario.
- (e) What are the necessary properties for choosing a suitable lubricant?
- (f) Explain crankcase ventilation.
- (g) What is meant by cylinder row and cylinder bank?
- (h) Name of various types of nozzles used in injection systems of IC engines.
- (i) Explain the factors affecting the combustion in IC engine.
- (j) What are the knocking and detonation in IC engines?

SECTION B

2. Attempt any three of the following:**10 x 3 = 30**

- (a) Explain stoichiometric ratio. How the stoichiometric ratio affect the heating value and emissions of an IC engine.
- (b) What are the various types of combustion chambers used in SI engines? Explain them briefly.
- (c) In a S.I. engine working on the ideal Otto cycle, the compression ratio is 5.5. The pressure and temperature at the beginning of compression are 1 bar and 27 °C respectively. The peak pressure is 30 bar. Determine the pressure and temperatures at the salient points, the air-standard efficiency and the mean effective pressure. Assume ratio of specific heats to be 1.4 for air.
- (d) What is the main function of a spark plug? Draw a neat sketch of a spark plug and explain its various parts
- (e) With a neat sketch explain a charcoal canister for controlling non exhaust emission.

SECTION C

3. Attempt any one part of the following:**10 x 1 = 10**

- (a) Explain the various types of fuel injection system used in CI engines. Which fuel injection system is suitable for CI engine? Justify your answer.
- (b) Explain turbo-charging. Give a brief note on variable geometry turbocharger and waste gate turbocharger.

4. Attempt any *one* part of the following:

10 x 1 = 10

- (a) Explain briefly
 - (i) mean effective pressure
 - (ii) fuel-air ratio
 - (iii) specific output
 - (iv) heating value of the fuel
 - (v) specific fuel consumption
- (b) Comparison of the Otto, Diesel and dual cycles on the basis of
 - (i) Same Compression Ratio and Heat Addition
 - (ii) Same Compression Ratio and Heat Rejection
 - (iii) Same Peak Pressure, Peak Temperature and Heat Rejection
 - (iv) Same Maximum Pressure and Heat Input

5. Attempt any *one* part of the following:

10 x 1 = 10

- (a) Explain the flame propagation in SI engines. Explain the various factors affecting flame propagation.
- (b) Explain the burning velocity of fuels. What are the factors affect it. Explain the method of measurement of burning velocity.

6. Attempt any *one* part of the following:

10 x 1 = 10

- (a) Explain CDI ignition system with a suitable diagram. Illustrate the advantages of CDI.
- (b) Explain the working of magneto ignition system. Differentiate between the magneto ignition system and battery ignition system.

7. Attempt any *one* part of the following:

10 x 1 = 10

- (a) Briefly explain the petroleum refining process
- (b) Write a brief note on engine emissions. What are the sources and factors affecting the emissions? Write short notes on Bharat Stage-VI for emission standard for vehicles.